Run ID: bier127_1791580517_d8f082

Adaptive Probe-Guided Genetic Algorithm (APGGS) — Experiment Report

Generated: 2026-10-10 02:45:31
Engine Version: v4.2

1. Executive Summary & Dataset Metadata

Dataset Name	BIER127	**Cities / Nodes**	127
Edge Weight Type	EUC_2D	**Known Best Optimum**	118,282
Final Global Best Cost	118,282.00	**Optimality Gap**	+0.0000%
Best Found Generation	15	**Completed Generations**	15
Stagnation Streak	0 / 30	**Stopping Condition**	30 gens without improvement
Experiment Status	COMPLETED	**Total Execution Time**	8.96 s
Stop Reason	0.0% gap achieved (Known optimum found)	**Random Seed**	1

2. Initial High-Quality Route Construction

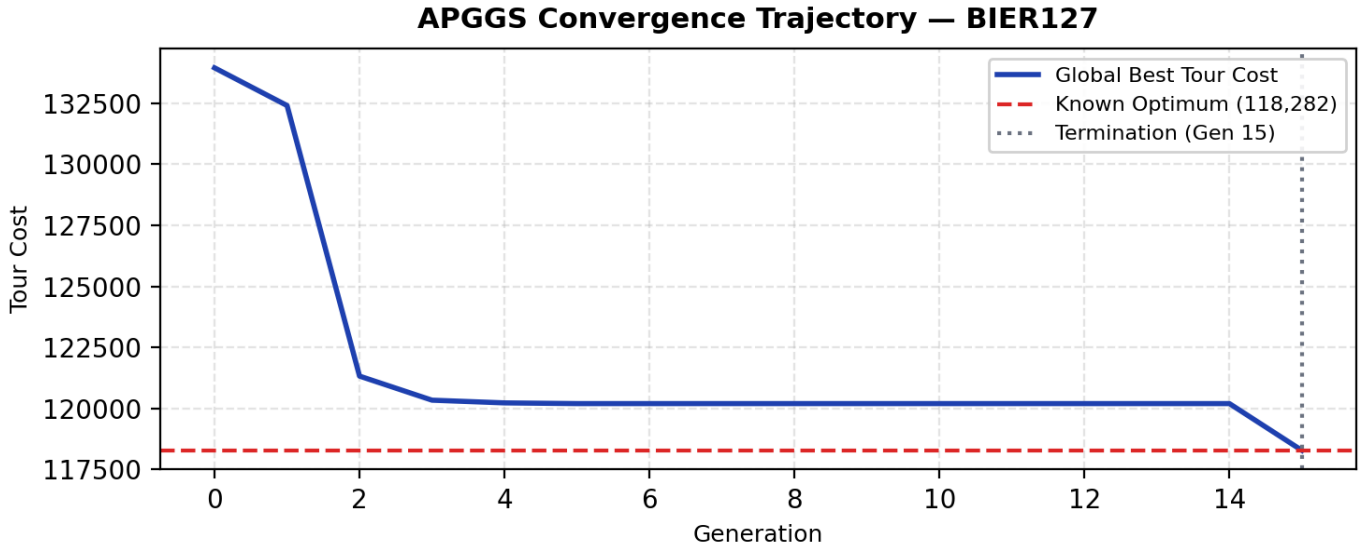
Construction Stage	Resulting Tour Cost	Improvement vs Prev	Duration (s)
Stage 1: Greedy Nearest-Neighbor Tour	141,339.00	Baseline	0.0148
Stage 2: 2-Opt Local Search Improvement	141,176.00	-163.00	0.0002
Stage 3: 2-Opt Deep Refinement Seed	134,293.00	-6,883.00	0.0008
Stage 4: Initial 70-Individual Population (NN+Greedy+Random)	133,953.00	Population Seeded	0.6500

3. Genetic Algorithm & Adaptive Pipeline Configuration

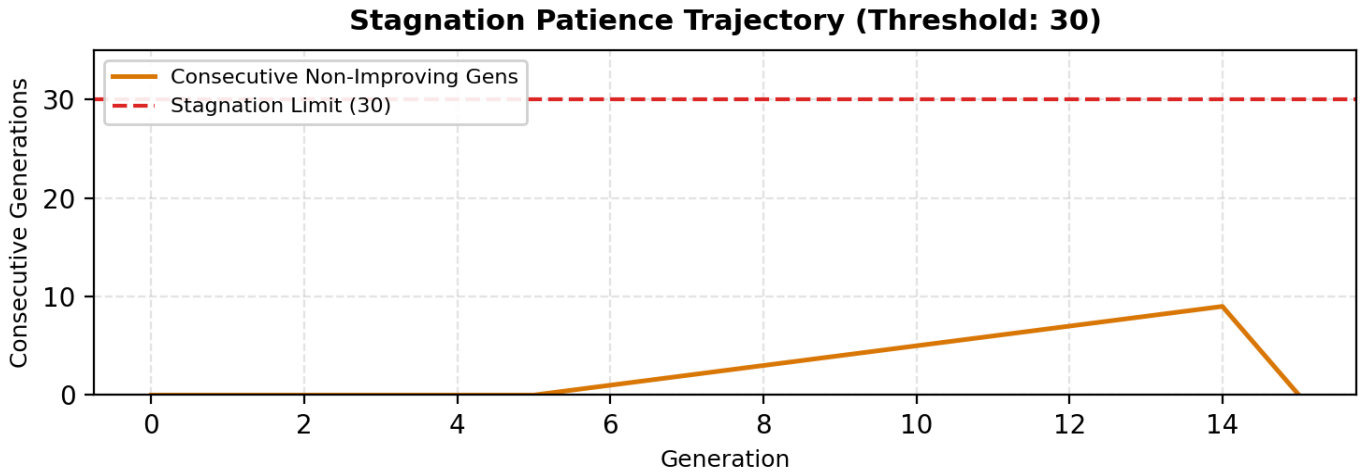
Parameter	Configured Value	Parameter	Configured Value
Population Size	70	Mutation Rate	0.20
Nearest-Neighbor Seeds	17	Swap Fraction	0.02
Greedy Edge Seeds	11	Segment Crossover Prob	0.40
Random Tour Seeds	42	Offspring 2-Opt Prob	0.35
K-Nearest Neighbors	15	Adaptive Elites Count	14
Tournament Size	4	PMG Memory Decay	0.92
Selective LKH Cadence	Every 15 gens	Stagnation Window	15
Selective LKH Individuals	1	Stagnation Patience Limit	30 consecutive gens

4. Research Visualizations & Trajectory Curves

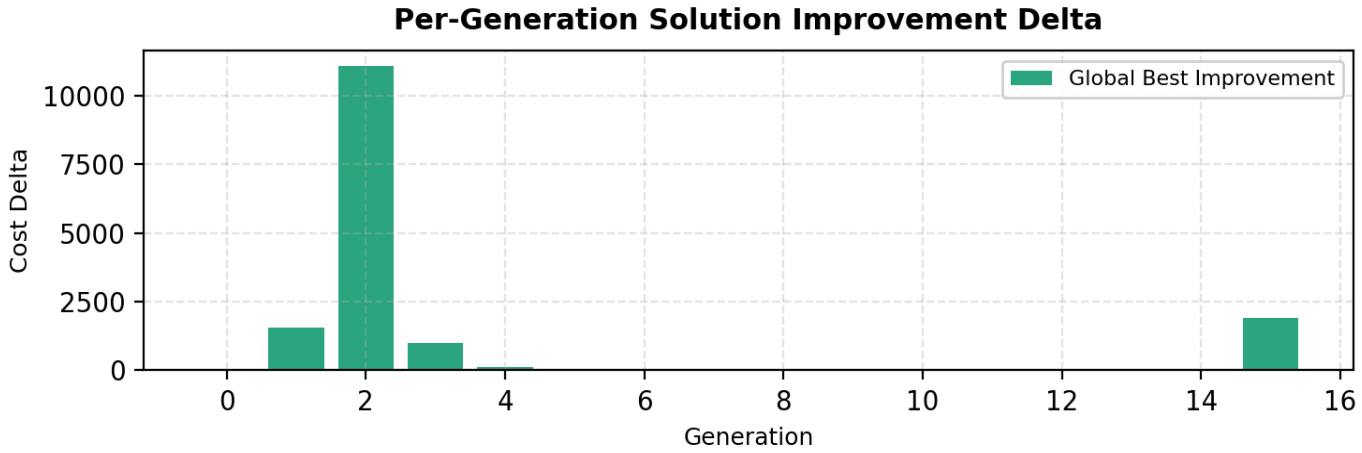
Convergence Trajectory: Global Best Cost vs Generation

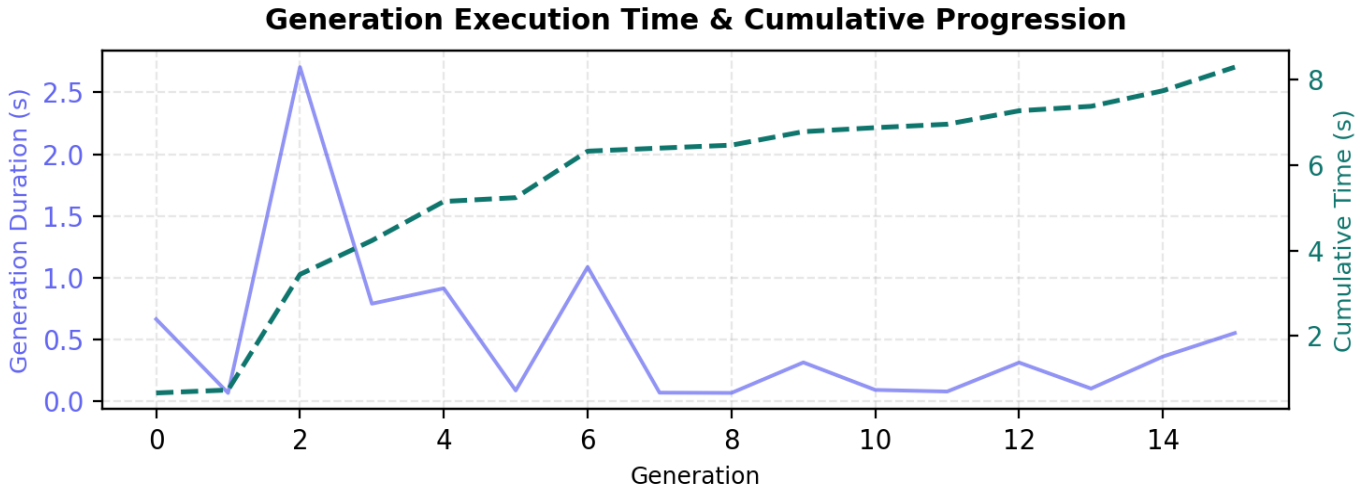


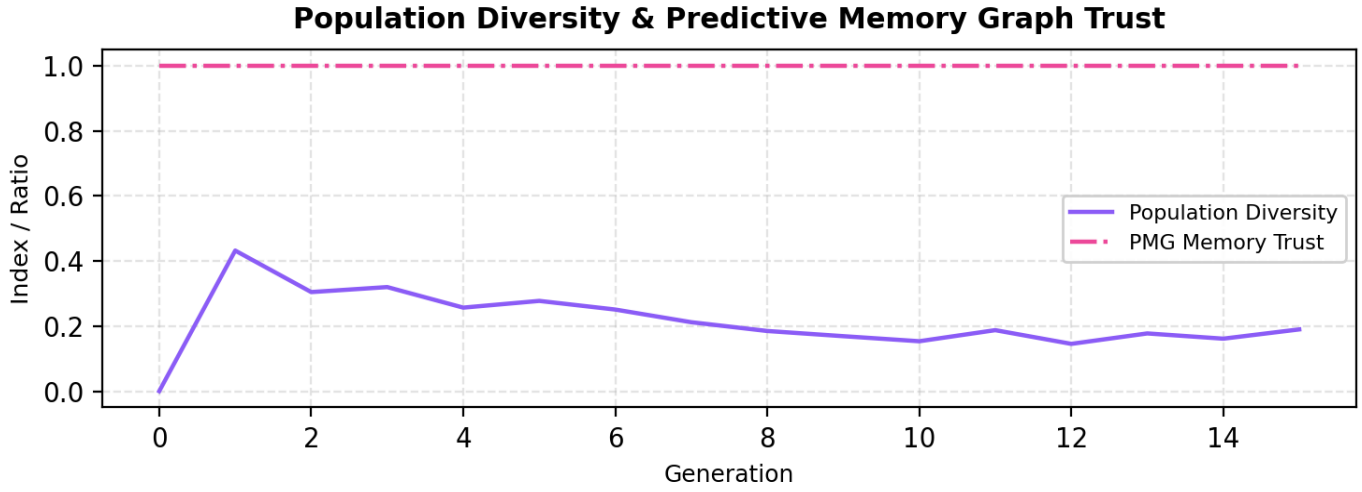
Stagnation Counter Progression (Target: 30 Consecutive Non-Improving Gens)



Per-Generation Cost Improvement & Runtime Progression







5. Evolution Highlights (Global-Best Discoveries)

Gen	Global Best Cost	Cost Reduction	Gap (%)	Operator / Method	PMG Trust	Non-Imp Streak	Duration (s)
0	133,953.00	0.00	+13.2488%	Nearest Neighbor (start=116)	1.00	0	0.6657
1	132,409.00	1,544.00	+11.9435%	Elite + two_opt	1.00	0	0.0689
2	121,326.00	11,083.00	+2.5735%	OX Crossover + predictive_or_opt	1.00	0	2.7051
3	120,340.00	986.00	+1.7399%	OX Crossover + Fast 2-Opt + predictive_or_opt	1.00	0	0.7912
4	120,229.00	111.00	+1.6461%	OX Crossover + Fast 2-Opt + predictive_or_opt	1.00	0	0.9153
5	120,200.00	29.00	+1.6215%	Elite + two_opt	1.00	0	0.0888
15	118,282.00	1,918.00	+0.0000%	Elite + Selective LKH	1.00	0	0.5535

6. Final Discovered Tour & Solution Vector

Optimized Tour Length: 118,282.00 | **Optimality Gap:** +0.0000% | **Discovered at Generation:** 15

Permutation Sequence Preview:

```
0 → 6 → 104 → 113 → 5 → 105 → 14 → 107 → 19 → 16 → 20 → 17 → 76 → 78 → 79 → 30 → 26 → 29 → 11 → 13 → 40 → 35 → 36 → 34 → 39 → 42 → 33 → 41 → 38 → 37 → 25 → 24 → 32 → 121 → 27 → 28 → 31 → 97 → 96 → 122 → ... [87 additional cities] → 0
```

7. Execution Event Log & Diagnostics

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• Initialized bier127 (127 cities). Initial best cost: 133,953.00
• Gen 1: Discovered new global-best 132,409.00 (-1,544.00)
• Gen 2: Discovered new global-best 121,326.00 (-11,083.00)
• Gen 3: Discovered new global-best 120,340.00 (-986.00)
• Gen 4: Discovered new global-best 120,229.00 (-111.00)
• Gen 5: Discovered new global-best 120,200.00 (-29.00)
• Gen 15: Discovered new global-best 118,282.00 (-1,918.00)
• Gen 15: 0.0% optimality gap achieved. Experiment completed.
```

8. Artifact Manifest & Research Reproducibility

All raw experimental records have been saved to runs/bier127_1791580517_d8f082/ including summary.json, generation_history.jsonl, best_route.csv, and report.pdf.